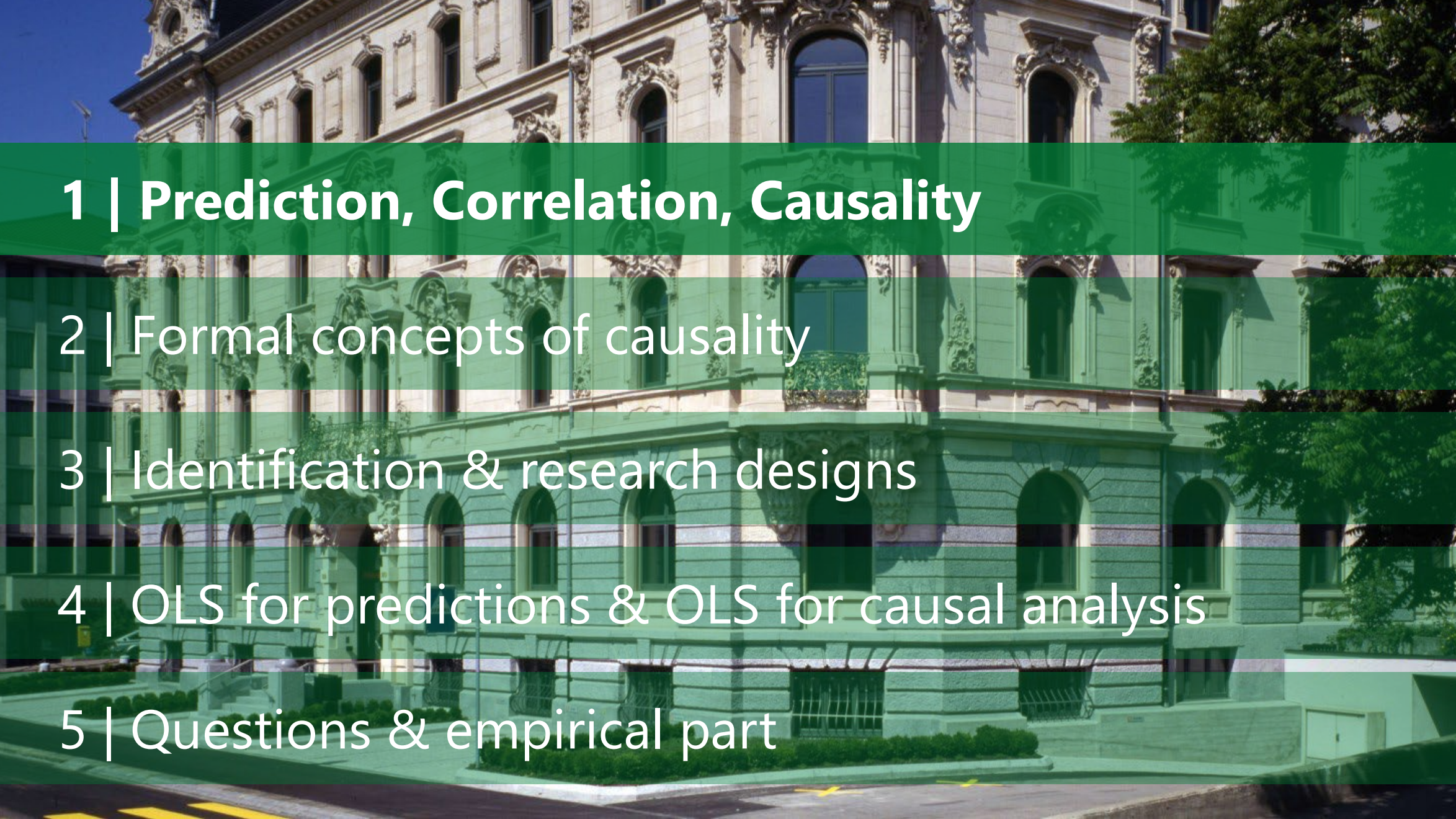
Causal Machine Learning & Artificial Intelligence for Economists

Michael Lechner
Swiss Institute for Empirical Economic Research (SEW-HSG)
University of St. Gallen | Switzerland

Lecture 1.2: Prediction & Causality

Required external Python modules (from PyPI): statsmodels

Disclaimer: Almost all figures created by ChatGPT



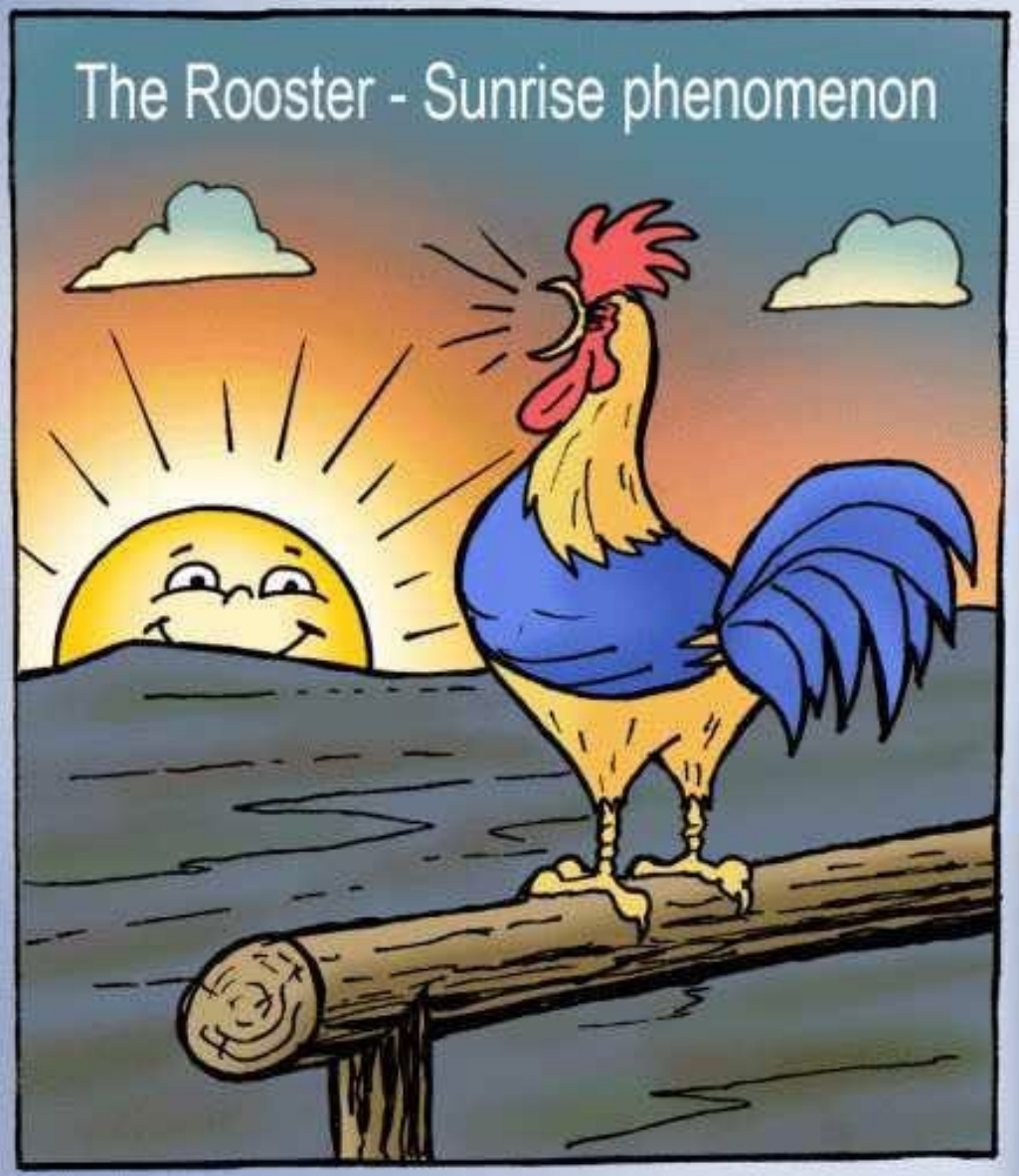
1 | Prediction, Correlation, Causality

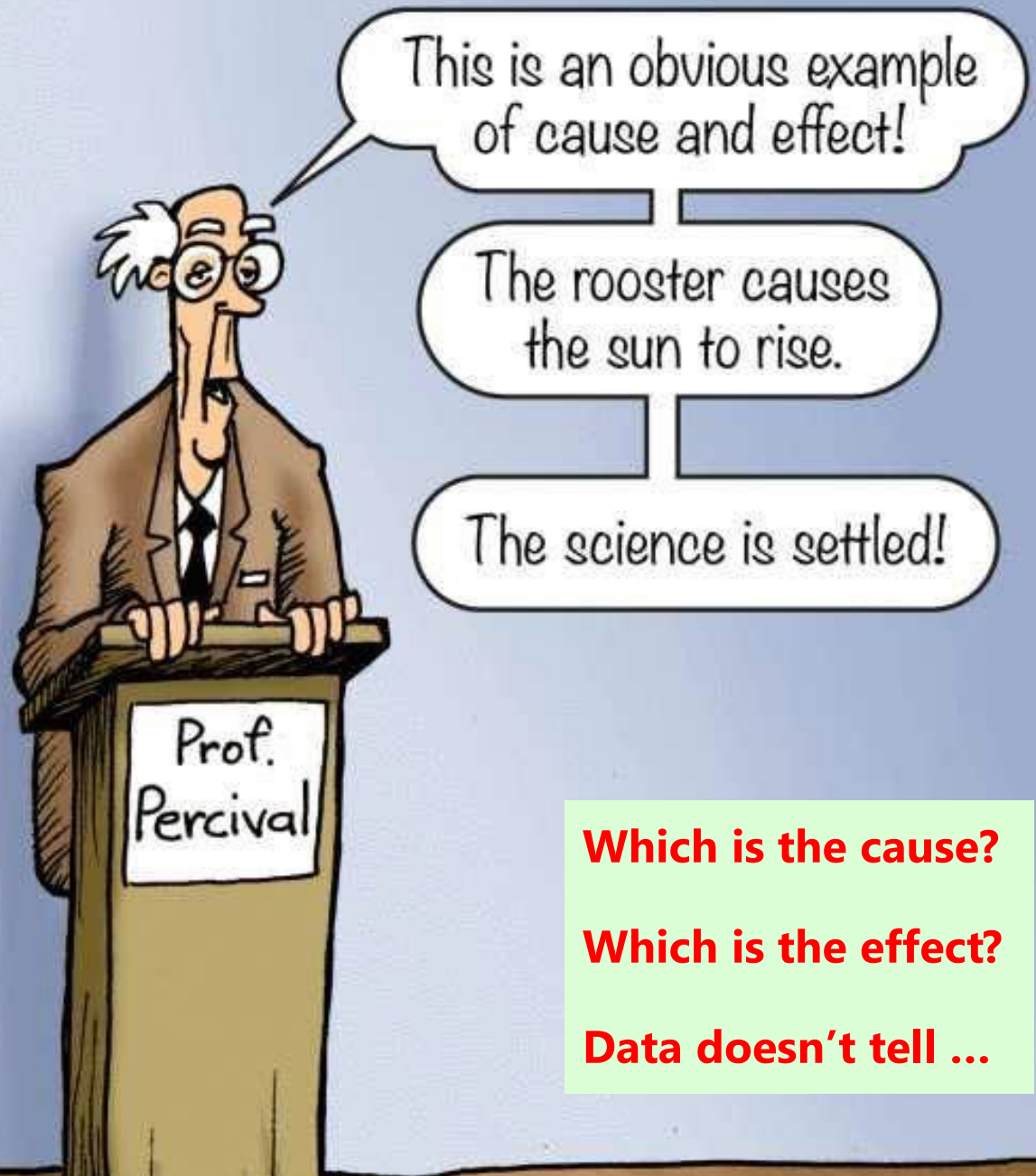
2 | Formal concepts of causality

3 | Identification & research designs

4 | OLS for predictions & OLS for causal analysis

5 | Questions & empirical part





Which is the cause?

Which is the effect?

Data doesn't tell …









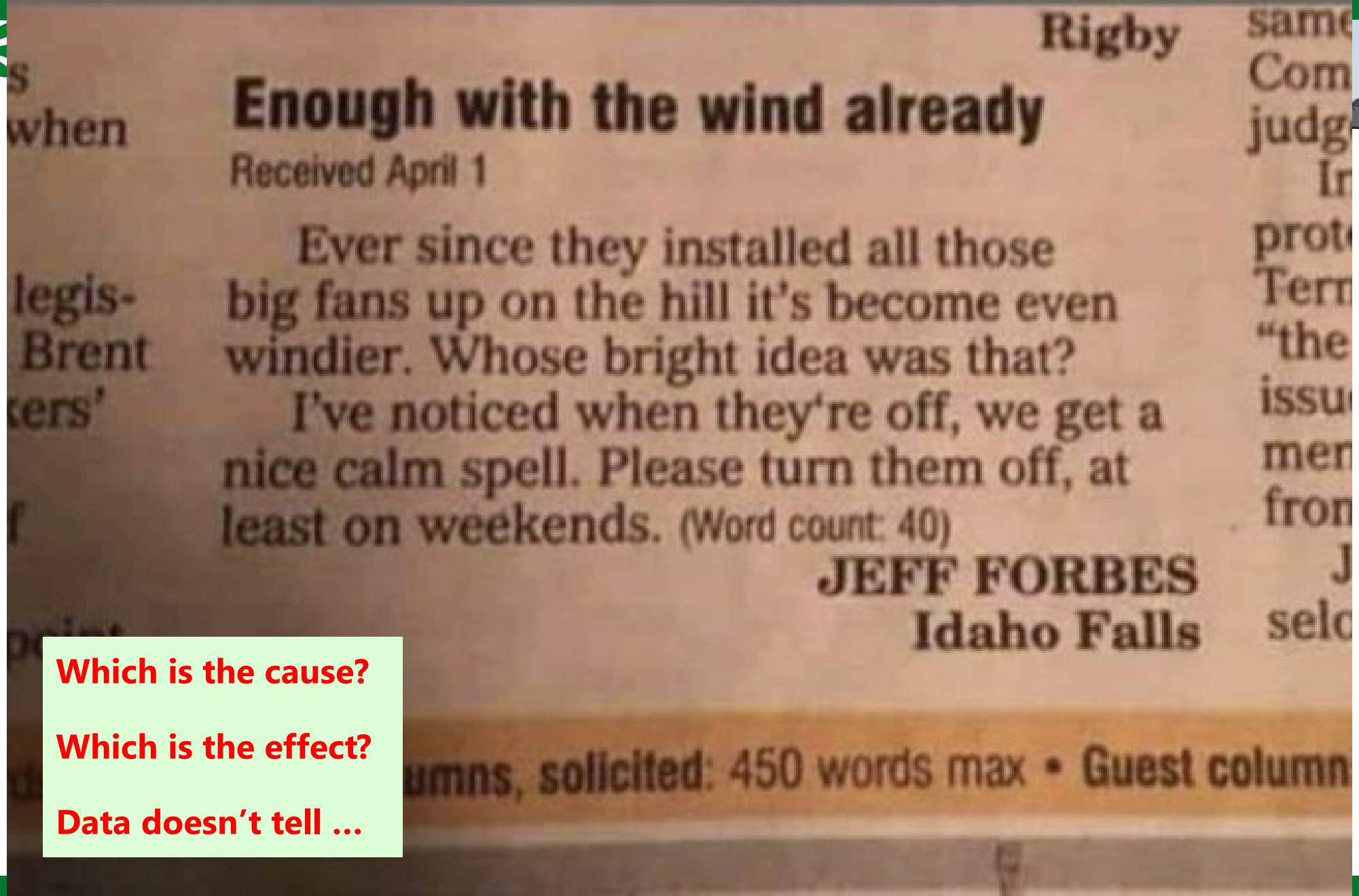


Maybe, the crew should not turn the seat belt sign on so often, because every time they do, it gets bumpy.

Which is the cause?

Which is the effect?

Data doesn't tell …



Rigby

Enough with the wind already

Received April 1

Ever since they installed all those big fans up on the hill it's become even windier. Whose bright idea was that?

I've noticed when they're off, we get a nice calm spell. Please turn them off, at least on weekends. (Word count: 40)

JEFF FORBES
Idaho Falls

umns, solicited: 450 words max • Guest column

Which is the cause?

Which is the effect?

Data doesn't tell …

University of St.Gallen

Swiss Institute for
Empirical Economic Research



John List (He/Him) • 1.
Distinguished Service Professor at University of Chicago
2 Wochen •

A High School student just wrote to me asking for help on discerning correlation and causation. They said that they are visual learners. I sent this picture along.

Übersetzung anzeigen



Sie und 11 weitere Personen

To understand cause → effect we need the full picture



Correlation & causal effects may have different signs

Suppose *D* causes *Y*

- No reason that *corr(D,Y)* has same sign as *effect*(*D* → Y)
- Famous example for how easy it is to have a sign change: *Simpson's paradox* (next)



A numerical example of *Simpson's paradox*

New Support Programme for Children in Need *HelpKids25* is tested in a trial

- 40 boys participated: 10 got support, 30 got nothing $P(support=1|boy)=0.25$
- 40 girls participated: 30 got support, 10 got nothing $P(support=1|girl)=0.75$
- For boys as well as for girls, getting the support of *HelpKids25* is random
- Outcome is measured as *completing high school* $P(HS\ completed|boys) > P(HS\ completed|girls)$

Mean comparisons reveal that *HelpKids25*

- … increases high school completion rate for girls
- … increases high school completion rate for boys
- … decreases high school completion rate overall

How is this possible? Does *HelpKids25* work, or not?



Girls	High School completed	High School not completed	Sum	High School completion rate
HelpKids25	9	21	**30**	**30%**
Nothing	2	8	**10**	**20%**
All girls	11	29	**40**	27%
Boys				
HelpKids25	7	3	**10**	**70%**
Nothing	18	12	**30**	**60%**
All boys	25	15	**40**	63%
Girls & boys				
HelpKids25	16	24	**40**	**40%**
Nothing	20	20	**40**	**50%**
All	36	44	**80**	45%

Lesson: A comparison is causal only within groups where treatment assignment is as good as random.

Artificial example | Simpson's paradox | Explanation

The bias in the overall mean comparison is due to **confounding** (*selection bias*)

- Boys are less likely to receive support than girls
- BUT: Boys are more likely to complete HS than girls (without support)

Gender is a *confounder*!

True population effect is an increase of HS completion by 10%-points

- (Mean effect of boys x population share of boys) + (Mean effect of girls x population share of girls):
  0.1 x 0.5 + 0.1 x 0.5 = 0.1
- *Non-data knowledge* used for this calculation
  - Data is from experiment stratified by gender

Weighted average of effects in unconfounded subpopulations **instead of mean comparison in full population!**

HelpKids25 **works nicely**

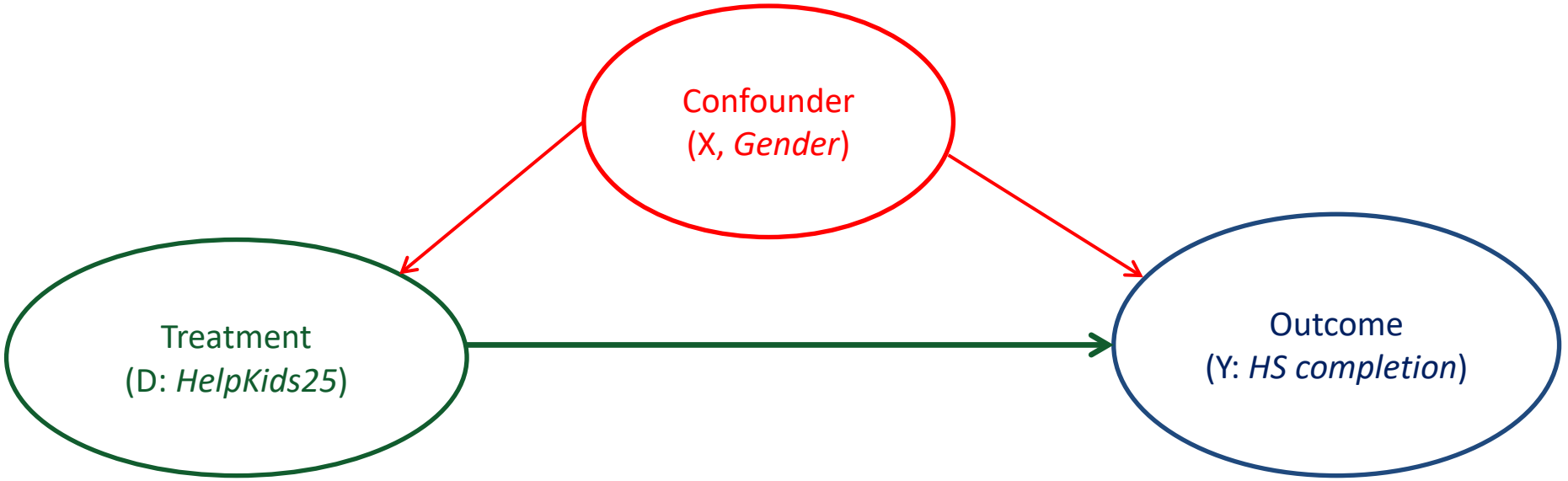
- Mean comparison in full population is misleading, because of *selection bias (ignoring confounder)*



Confounders

A confounder is a variable that *confounds* the comparison of treatment & outcomes ➔ selection bias

- More formal definition will follow





Challenge of causal analysis

How to deal with observed & unobserved confounders to adjust associations such that they reveal causal effects?

Conceptual framework: Counterfactual worlds

- What would be the value of the outcome if treated?
- What would be the value of the outcome if not treated?



Example from *rainonomics*

Predictive vs. causal questions

Q1: Shall I do a rain dance to increase likelihood of rain?

- Causal problem (estimate effect of rain dance on rain), because performing a rain dance (might) influence whether it rains or not

Q2: Shall I take an umbrella with me when I leave home?

- Predictive problem (estimate likelihood of raining), because taking an umbrella does not influence whether it rains or not, but knowing whether it rains or not is most valuable for this decision

Prediction & causation are different concepts which have different uses

This example is taken from my colleague Jens Ludwig (Chicago).







Predictive & causal questions | The unemployment rate

Predictive: For next year's budget of its employment services, the government needs an estimate of the unemployment rate to estimate the outlays in employment benefits

Causal: The government considers an increase in unemployment benefits by 5%. It wants to know how the unemployment rate is affected by such a potential policy change

Prediction vs. Causal Analysis

Two different questions about unemployment require two different calculations

Q1: Predictive question

For next year's budget, the government needs an estimate of the unemployment rate.

Observed and predicted unemployment rate

(Percent of labor force)



Goal: predict what will happen next

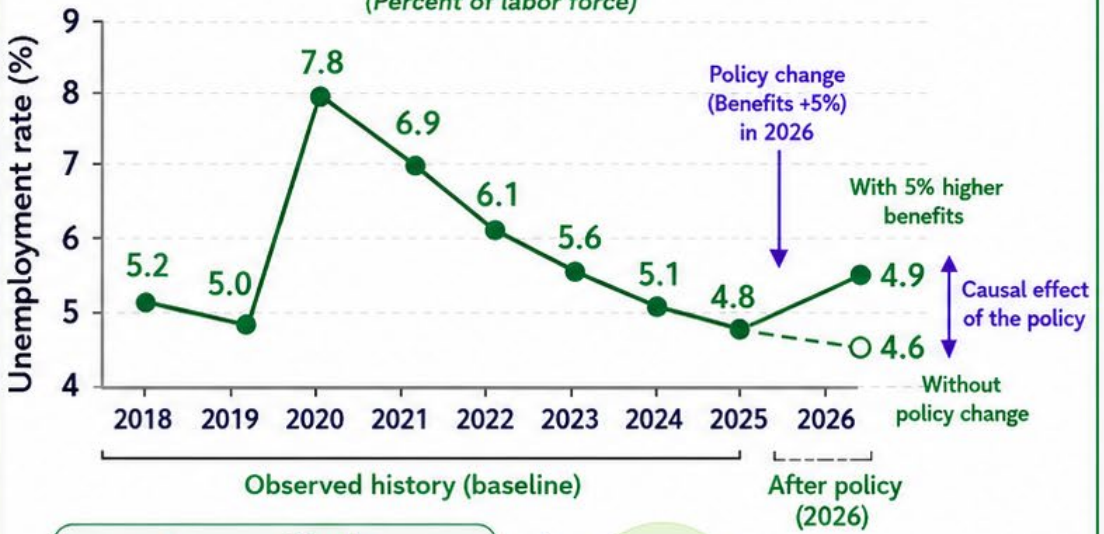
Uses historical patterns and predictors

Q2: Causal question

What happens to unemployment if benefits are increased by 5%?

Observed and predicted unemployment rate

(Percent of labor force)



Goal: estimate what would happen under a different policy

Requires a counterfactual comparison

Prediction asks: What will unemployment be? | **Causal analysis asks:** How would unemployment change if policy changed?



Predictive & causal questions | Targeting vs Profiling

There are two possible strategies on how to use individual data to decide which programme is best for a specific unemployed person

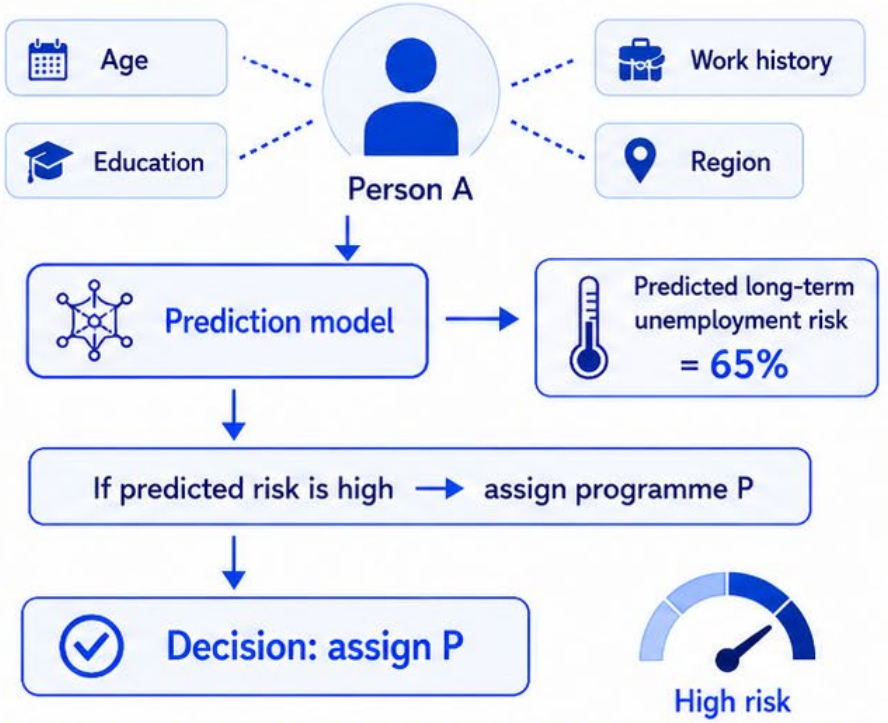
- **Profiling:** allocate support based on predicted risk:
  "Who is likely to become long-term unemployed?"
- **Targeting:** allocate support based on predicted treatment effects:
  "For whom does this programme reduce the risk of long-term unemployment?"

Profiling vs. Targeting

Two different ways to use individual data to decide which programme should be allocated

Profiling

Allocate programme P to person A if A has a high predicted risk of becoming long-term unemployed.



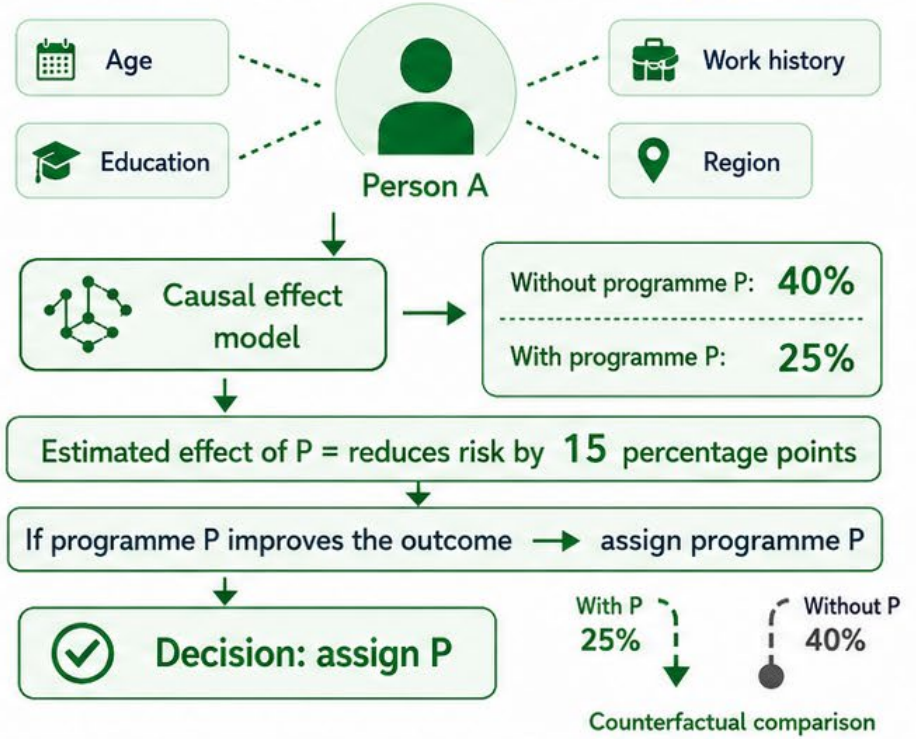
Uses a prediction of the outcome

Both use the same individual data about person A

But they ask different questions

Targeting

Allocate programme P to person A if programme P has a beneficial effect on A's long-term unemployment risk.



Uses an estimate of the treatment effect

Profiling asks: *Who is at high risk?*

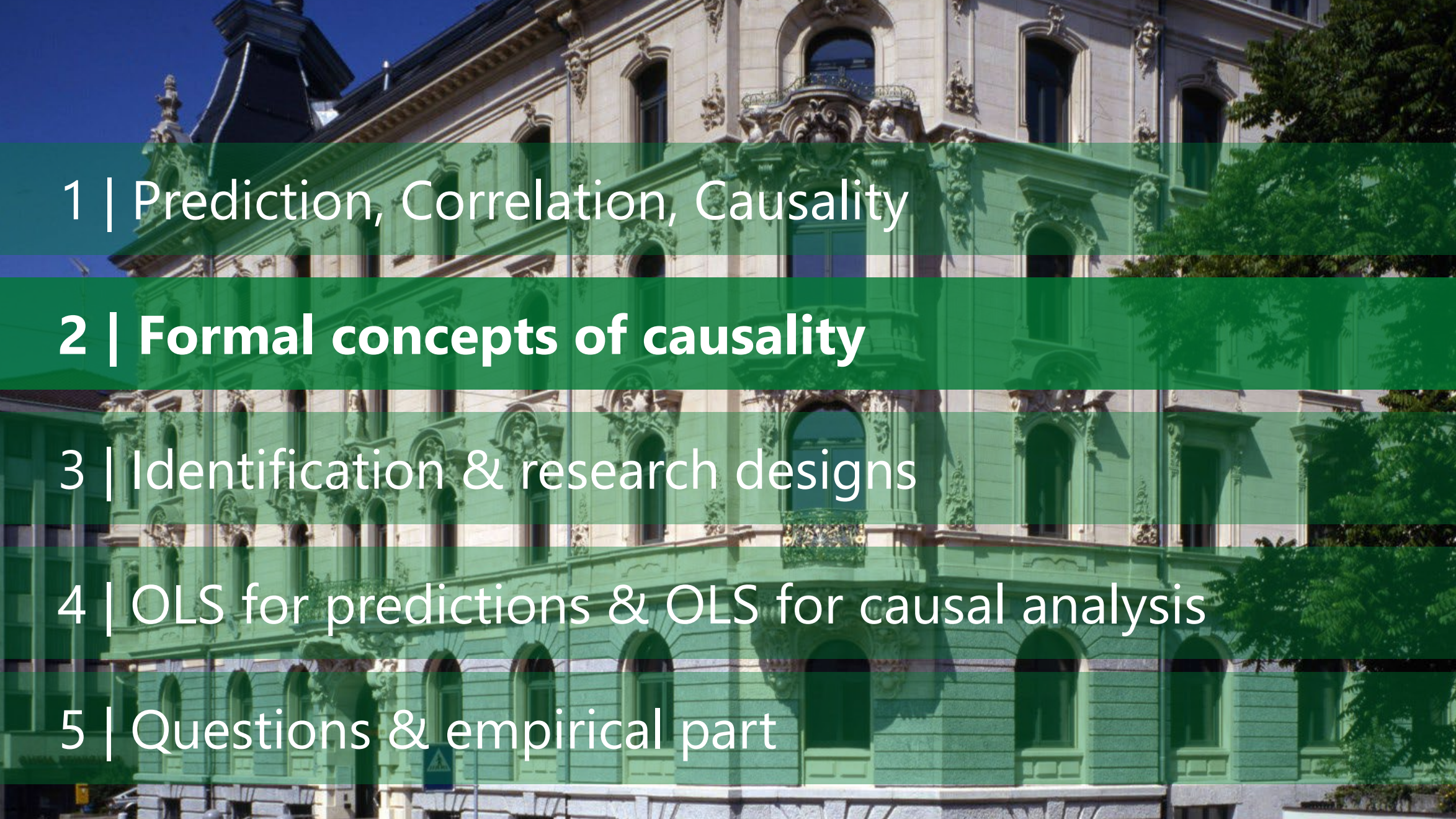
Targeting asks: For whom does programme P work?



Next steps

Intuitively, we think we 'know' the difference between causality / correlation / prediction

However, for a clear empirical analysis, we need a formalisation of it → **Next**



1 | Prediction, Correlation, Causality

2 | Formal concepts of causality

3 | Identification & research designs

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Conceptual framework

Counterfactual worlds – ***what-if*** questions

- What would be the value of the outcome if treated?
- What would be the value of the outcome if not treated?



Two formal frameworks became popular

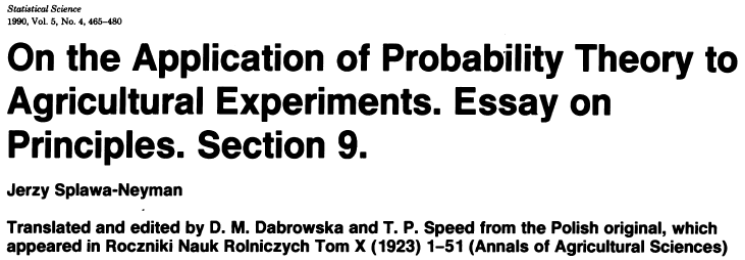
Potential outcomes (PO)

- Neyman (1923)
- Rubin (1974): Estimating causal effects of treatments in randomized and nonrandomized studies. *Journal of Educational Psychology, 66*(5), 688–701.

Directed Acyclic Graphs (DAG)

- Pearl, J. (2000), *Causality - Models, Reasoning, and Inference*, Cambridge: Cambridge University Press

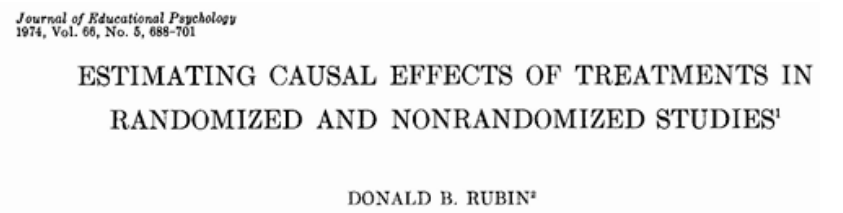
In many standard applications, both frameworks lead to the same empirical design


Statistical Science
1990, Vol. 5, No. 4, 465–480

On the Application of Probability Theory to Agricultural Experiments. Essay on Principles. Section 9.

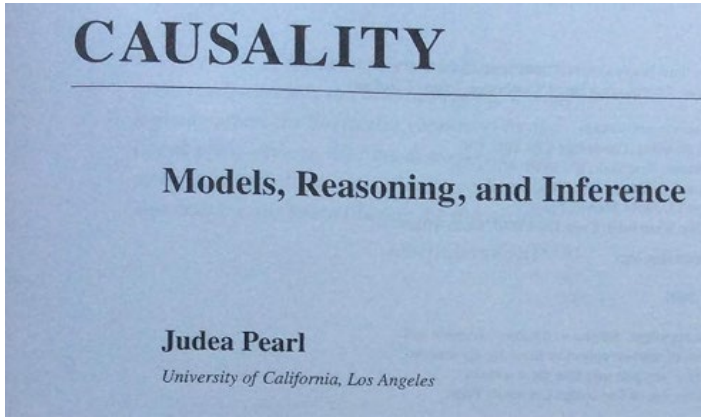
Jerzy Splawa-Neyman

Translated and edited by D. M. Dabrowska and T. P. Speed from the Polish original, which appeared in Roczniki Nauk Rolniczych Tom X (1923) 1–51 (Annals of Agricultural Sciences)


Journal of Educational Psychology
1974, Vol. 66, No. 5, 688-701

ESTIMATING CAUSAL EFFECTS OF TREATMENTS IN RANDOMIZED AND NONRANDOMIZED STUDIES[1]

DONALD B. RUBIN[2]


CAUSALITY

Models, Reasoning, and Inference

Judea Pearl

University of California, Los Angeles



Notation of the potential outcome model

We need a notation that reflects the *counterfactuals*

Treatment:	D	(for simplicity: 0, 1)	observable
Observed outcome:	Y		observable
Features (covariates)	X		observable
Potential outcome when treated:	$Y(1)$	*observable if D=1, unobservable if D=0*	
Potential outcome when not treated:	$Y(0)$	*unobservable if D=1, observable if D=0*	

Connection of observable to potential outcomes: $Y = D\,Y(1) + (1 - D)\,Y(0)$



Definition of causal effects

Individual causal effect: Causal effect for a single unit

- *ITE = Y(1) - Y(0)*

Average causal effects: Averages of *ITE*s over groups

- Usual objects of interest in evaluation studies
- Average treatment effect: $ATE = E(Y(1)-Y(0))$
- Conditional average treatment effect: $CATE(x) = E(Y(1)-Y(0)|X=x)$

Causal Estimation = Solving a Missing Data Problem

For each individual, we only observe one potential outcome

Individual	Gender (X)	Y(0) earnings	Y(1) earnings	D	ITE = Y(1) - Y(0)
A	Female	40	?	0	? - 40
B	Male	?	55	1	55 - ?
C	Female	48	?	0	? - 48
D	Male	?	62	1	62 - ?
E	Female	52	?	0	? - 52
F	Male	?	58	1	58 - ?

Red ? = unobservable counterfactual outcome

Each person has two potential outcomes, Y(0) and Y(1), but we only observe the one corresponding to the actual treatment status D. The other potential outcome is missing.



Challenge of causal analysis (see Simpson's paradox)

How to adjust associations/correlations for observed & unobserved confounders so that they reveal causal effects?

These adjustments depend on our expert knowledge on the relation of confounders, potential outcomes, & treatments

➔ ***Next: Identification / research designs***



1 | Prediction, Correlation, Causality

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Definition of identification

A parameter is identified if it can be expressed in terms of random variables for which observations can be sampled

(or: … identified if we can learn it from the observable data distribution)

- This is usually unproblematic for prediction problems, but not for causal problems (because of the counterfactual)



Identification of causal effects require additional assumptions

Without further assumptions, ATE is ***not*** identified

ATE = E(Y(1)-Y(0))

- *Observations of Y(1) can only be sampled for observations with $d_i=1$*
- *Observations of Y(0) can only be sampled for observations with $d_i=0$*



How experiments solve the causal problem

If D is randomized ➔ *Y(1), Y(0)* is independent of *D* ➔

- distribution of *Y(1)* is the same for treated & non-treated
- distribution of *Y(0)* is the same for treated & non-treated

Identification idea

- Take observations of *Y* of the non-treated to approximate what would have happened to the treated if NOT treated
- Take observations of *Y* of the treated to approximate what would have happened to the non-treated if treated

➔

$$ATE = E(Y \mid D = 1) - E(Y \mid D = 0),$$

$$CATE(x) = E(Y \mid D = 1, X = x) - E(Y \mid D = 0, X = x)$$



Observational studies

No experimental data available

- Try to use the data such that the results are (almost) as credible as coming from an experiment
- Causal analysis by quasi-experimental methods
- Different quasi-experimental methods are based on different assumptions about the world (= research designs)
- Examples: Instrumental Variables, Difference-in-differences, Unconfoundedness, Regression-discontinuity-design, …

In this course, we will focus on so-called '*Unconfoundedness*'



Unconfoundedness | 1

Assume data is *as good as* coming from many little experiments given all possible values of *X* (stratified experiments) ➔ *Y(1), Y(0)* is independent of *D for each value x of X*

Assume that *X* is not a perfect predictor of *D* (*common support):* $0 < P(D = 1 \mid X = x) < 1, \forall x \in \mathrm{X}$

Identification idea

- Take observations of *Y* of the non-treated *with* x_i=*x* to approximate what would have happened to the treated *with value x* if not treated
- Take observations of *Y* of the treated *with value* x_i=*x* to approximate what would have happened to the non-treated *with value x if* treated
- $CATE(x) = E(Y \mid D = 1, X = x) - E(Y \mid D = 0, X = x)$
- $ATE = E_X CATE(x) = E_X E(Y \mid D = 1, X = x) - E_X E(Y \mid D = 0, X = x)$



Unconfoundedness | 2

Unconfoundedness allows treated & non-treated to have different distributions of X

- We will remove these differences by statistical methods (e.g., Causal Machine Learning)

Unconfoundedness only holds if

- All confounders are observed (& included in *X*) (**C**onditional **I**ndependence **A**ssumption)
  - non-statistical knowledge needed to understand which variables are confounders
- *X* is *not* a perfect predictor of treatment (**C**ommon **S**upport)
- The measurement of *X* is not affected by the treatment (**E**xogeneity)



1 | Prediction, Correlation, Causality

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Linear Regression Model & OLS revisited | 1

Notation of linear regression

$$Y = D\alpha + X\beta + U = Z\gamma + U, \quad Z = (D, X), \quad \gamma = (\alpha, \beta)$$

For very large samples, OLS estimators will converge to

$$\hat{\gamma}_\infty = E(Z'Z)^{-1}EZY, \qquad \hat{\gamma}_\infty = \lim_{N\to\infty} \hat{\gamma} \qquad \text{(coefficients)}$$

$$\hat{Y}_\infty(z) = z'E(Z'Z)^{-1}EZY = \hat{\gamma}_\infty z, \quad \hat{Y}_\infty(z) = \lim_{N\to\infty} \hat{Y}(z) \qquad \text{(target of prediction)}$$

$$\hat{\alpha}_\infty = [E(Z'Z)^{-1}]_{00}EDY, \qquad \hat{\alpha}_\infty = \lim_{N\to\infty} \hat{\alpha} \qquad \text{(target of causal analysis)}$$



Linear Regression Model & OLS revisited | 2

What is needed for a good **prediction** with OLS?

- Estimator is computable (➔ no multicollinearity)
- Good out-of-sample fit (high R^2, low **M**ean **S**quared **E**rror, etc.)

What is needed for **causal analysis** with OLS?

- Assumptions of the Linear Regression **Model** must be a correct description of the **D**ata **G**enerating **P**rocess
  - ***E(U|X=x, D=d)=0, for all values of x, d***, no multicollinearity

    ➔ Unconfoundedness, effect homogeneity & linearity of $E(Y|D=d, X=x) = d\alpha + x\beta$



Prediction & causality | Summary

	Prediction	Causal analysis
Question	What will happen?	What would happen under a policy change?
Object	Predicted outcome	Causal effect
Notation	$E(Y \mid X)$, $P(Y = 1 \mid X)$	$Y(1)$, $Y(0)$, ATE, CATE(x)
Main challenge	Good out-of-sample prediction	Credible identification
Success criterion	Low prediction error	Convincing causal interpretation
Example	Will it rain?	Does a rain dance increase rain?

Same data may answer different questions: prediction needs fit; causality needs identification.



1 | Prediction, Correlation, Causality

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Rules for group work (including graded group exercise)

Groups may have up to 4 members

- *All group members must agree to all group members*

Group memberships may change until lecture 3

You are free to use any AI you like

- Graded exam: Contact with humans is prohibited & will be monitored (& will be useless )

Lecturer is available for discussions & help during group work

- limited in graded group work

You should always ***install & test*** the required packages ***before the lecture***

- ***Required (additional) packages are always shown on 1st slide of each lecture***



Empirical group work | General structure

Goal is to show the potential & limits of regression for prediction & causal analysis

The task consist of the following steps:

1) Generate 2 data sets (T, P) of size *N* for *Y, D, X* using 4 different **D**ata **G**enerating **P**rocesses. Use *T* for training & *P* for out-of-sample prediction & validation
2) Run a prediction with OLS for the different DGPs (***statsmodels***, self-written OLS, scikit-learn, …)
3) Estimate the causal effect of *D* on *Y* for the different DGPs using different specifications
4) Compare performances across tasks, DGPs & specifications (bias, RMSE) for very large samples



Empirical group work | Details | DGPs (1)

Consider the following 4 different DGPs

$$y_i = 1 + d_i\alpha_0 + (x_i\beta_0)^{\gamma_0} + u_i$$
$$d_i = \underline{1}(x_i\lambda_0 + v_i > 0), \quad \forall i = 1,\ldots,N_T, N_T+1,\ldots,N_T+N_P$$
$$u_i \sim N(0,1), \quad v_i \sim N(0,1), \quad x_i \sim N(0,1), \quad iid \text{ draws}$$
$$\alpha_0 = 1, \quad \beta_0 = 1, \quad N_T, N_P \geq 100'000$$
$$\lambda_0 \in (0,5), \quad \gamma_0 \in (1,2) \qquad \text{(4 different DGPs)}$$



Empirical group work | Details | Prediction (2)

1) For each of the 4 DGPs, train an OLS on data *T* only

2) Compute predictions of *Y* for *T* & *P* (using OLS coefficients obtained from *T* only)

3) Compute *the **R**oot **M**ean **S**quared Error* of the predicted *Y* for *T* & *P*

Do this for the following 2 specifications of explanatory variables in OLS

- constant, *D*, *X*
- constant *D*, *X*, X^2

Which DGP & OLS-specification combination gives the best / ok / worst out-of-sample results?

- Explain these performance difference by how the specifications match to the DGPs



Empirical group work | Details | Causal analysis (3) & comparison (4)

Use same DGPs & specification of OLS as before

Compute the differences of the true & estimated causal effect

Which DGP & OLS-specification combination gives the best / ok / worst results?

- Explain these performance difference by how the specifications match to DGPs

Compare the quality of the findings for prediction & causal analysis

Bonus for the fast groups: Repeat these simulations for N_T=10 & N_T=10'000'000. Do your findings change?



Further (easy) reading

SPECIAL SECTION PREDICTION

ESSAY

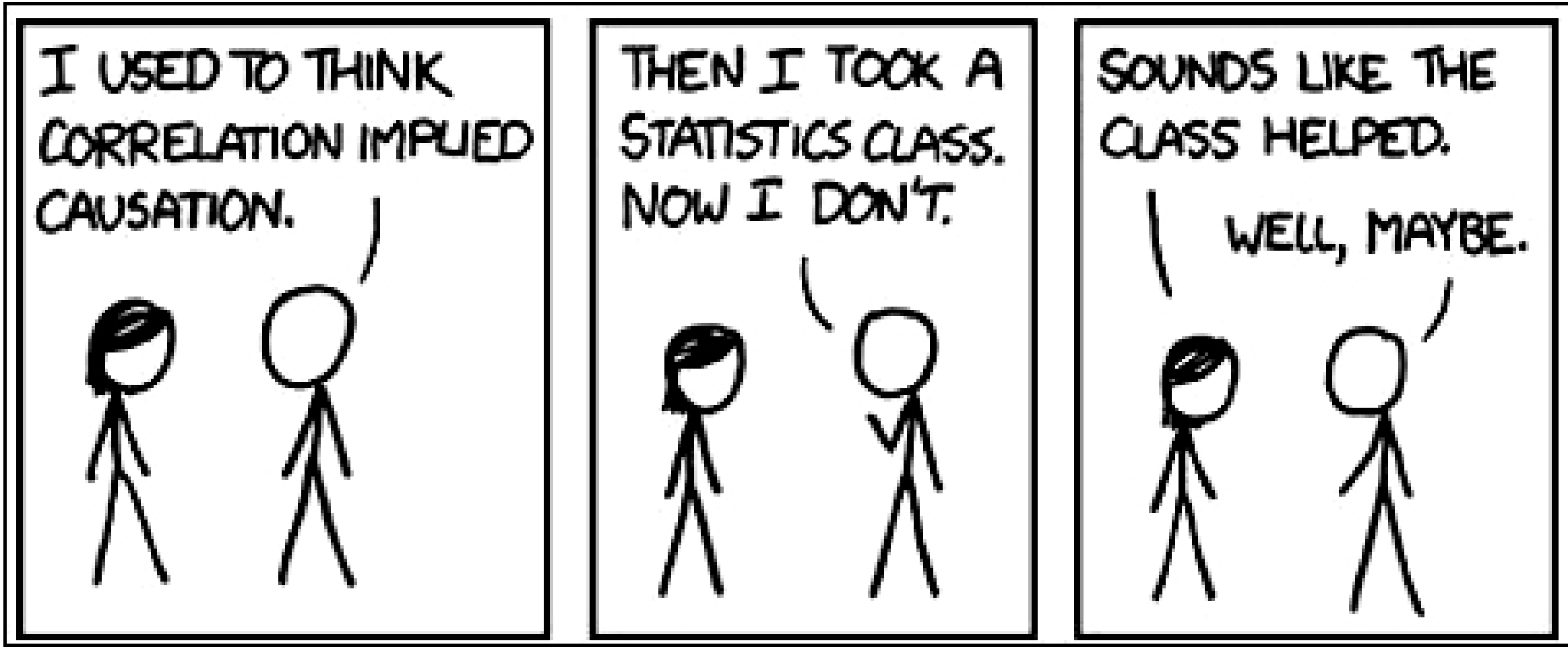
Beyond prediction: Using big data for policy problems

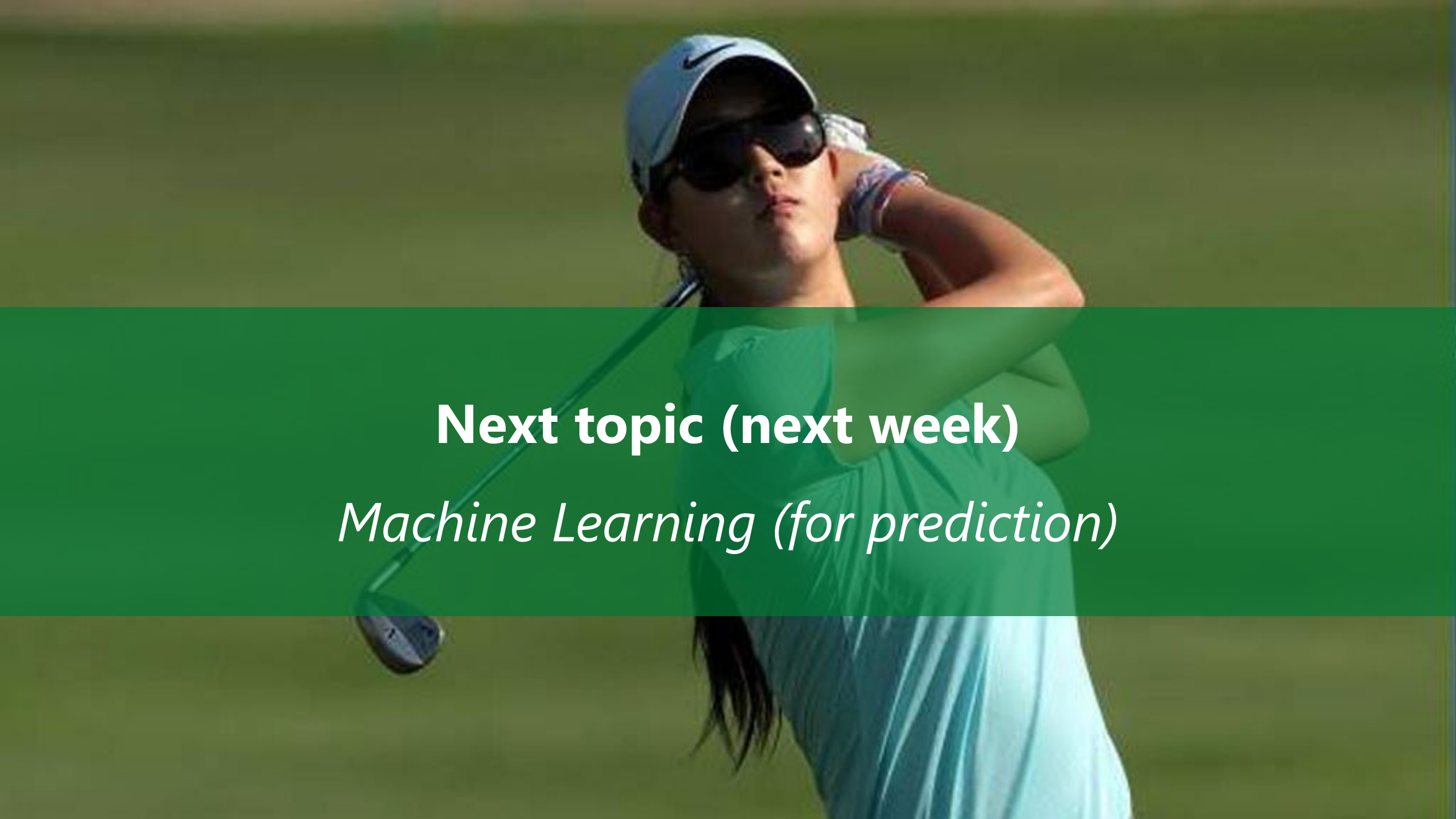
Susan Athey

Machine-learning prediction methods have been extremely productive in applications ranging from medicine to allocating fire and health inspectors in cities. However, there are a number of gaps between making a prediction and making a decision, and underlying assumptions need to be understood in order to optimize data-driven decision-making.

Athey, *Science* **355**, 483–485 (2017) 3 February 2017




I USED TO THINK CORRELATION IMPLIED CAUSATION.
THEN I TOOK A STATISTICS CLASS. NOW I DON'T.
SOUNDS LIKE THE CLASS HELPED.
WELL, MAYBE.



Next topic (next week)

Machine Learning (for prediction)